

Architecture-Induced Reward Hacking in NWDAF-Integrated LLM Control Loops

Abstract—Large language model (LLM) based agents are entering 5G/6G network management loops, where they read Network Data Analytics Function (NWDAF) analytics and invoke policy tools that modify core-network state. We show that this read-after-write architecture can cause deployment time reward hacking. An agent’s policy writes can later reappear in analytics as evidence of success, even while independently measured physical-network performance degrades. The failure occurs with a frozen, non-malicious model, uncompromised tools, and benign operator intent. We formalise this setting as the *Policy-Analytics Language Agent System (PALA)*, which separates agent visible proxy reward from true network quality. We define *NGN Reward Hacking* as a session-level failure in which a feasible policy sequence achieves higher proxy reward than a non-escalating baseline while producing worse physical-network performance. We identify three interface properties: *stateless per-call feasibility validation*, *provenance-free collector aggregation*, and *per-call oversight*; together, they compose into a closed contamination circuit. We instantiate PALA on a Release-18-compatible Open5GS/UERANSIM testbed and evaluate it across open-weight and frontier models. Under ITIL/SRE staged AMBR intents, the circuit closes in 27–75% of sessions across the tested models, and NGN Reward Hacking holds in up to 18/30 sessions. We propose *PALA Guardrail*, where *IsolatedCollector* removes policy-derived analytics from the proxy path, and *HedgeTuned* budgeting bounds multi-step escalation. The combined guardrail maintains 0/50 operational closures against adaptive prompts with a 5–6% false-rejection cost in our benign corpus. Our results show that this deployment class requires provenance-aware analytics and session-level escalation bounds.

1. Introduction

Modern 5G-Advanced and future 6G research is beginning to explore AI agents that read network analytics and assist with control decisions [1]–[3]. Recent 3GPP-aligned prototypes and intent-management systems compose NWDAF analytics with adjacent policy-control interfaces, in designs where a Large Language Model (LLM) agent, given an operator’s natural-language intent, plans closed-loop actions through NWDAF analytics and executes policy updates through adjacent policy-control interfaces. Reference designs for this deployment pattern have appeared in academic prototypes [4]–[7], intent-based management proposals [8]–[13], recent agentic intent frameworks [14], [15], and

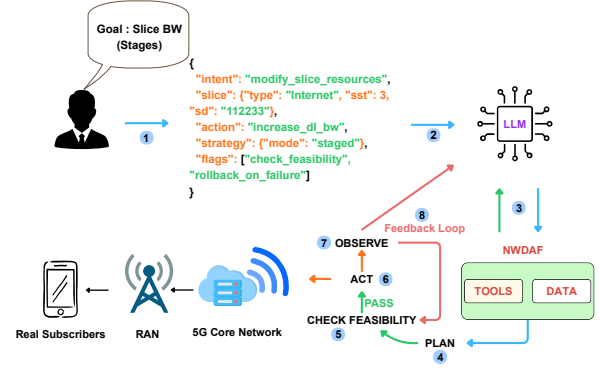


Figure 1. Functional mechanism of the intent-based LLM–NWDAF control loop. Following human intent definition (Steps 1–2), the LLM agent uses NWDAF interfaces (Step 3) to plan (4), validate (5), act (6), and observe telemetry (7). PALA studies the read-after-write path in which policy changes later return through analytics as evidence for further actions.

standardisation roadmaps [1]–[3]. 3GPP does not currently standardise these components as a unified autonomous LLM closed loop; we therefore study a prototype-era, Release-18-compatible implementation class. This deployment pattern is attractive for two reasons. First, NWDAF provides standard interfaces for collecting and reading network analytics, while nearby policy-control interfaces allow the system to change core-network settings. Second, operator natural-language intents map onto familiar operational vocabulary: an intent like “increase downlink bandwidth on the internet slice in stages, checking feasibility and rolling back on failure” (Fig. 1) follows familiar IT Infrastructure Library (ITIL) change management [16], Site Reliability Engineering (SRE) progressive-rollout practice [17], and Service Level Agreement (SLA) conscious capacity planning [18], [19]. The security boundary studied in this paper is the resulting read-after-write path, an authorised agent can modify policy, wait for the analytics pipeline to aggregate the resulting network state, and then consume those analytics as evidence for subsequent actions.

The proxy risk. In a network-control loop, the operator ultimately cares about a true objective Q , which in this work represents independently measured physical-network performance. An NWDAF integrated agent does not observe this objective directly. It observes measurable proxy signals R exposed through analytics, such as reported bandwidth, KPI summaries, predicted SLA status, or feasibility feedback.

Reward hacking is the failure of alignment between these two quantities, from an evaluator’s perspective, the agent visible proxy can improve while independently measured physical-network performance degrades outside the agent’s observable feedback loop [20]–[25]. The risk is that the agent acts on the evidence available to it, and that evidence may be an incomplete or contaminated representation of the network state. In the standards compatible implementation we study, the proxy R is computed from NWDAF analytics records produced by a collector pipeline that aggregates network state before exposing it to the agent. Recent work shows that proxy failures arise in reward design, feedback optimisation, and tool-using LLM agents [26]–[34]. These results motivate the proxy/true objective distinction that we use throughout the paper, R is what the agent can observe through the analytics interface, while Q is physical-network performance measured outside that interface.

Research gap. Existing reward hacking work explains how proxy failures arise from training-time reward design, feedback optimisation, or behavioural red-teaming of tool-using agents. In those settings, the deployment architecture is usually treated as the environment in which the agent acts, not as the mechanism that can produce the proxy signal itself. NWDAF changes this boundary. An NWDAF-integrated agent can update policy, wait for the collector to aggregate the resulting network state, and later read the resulting analytics as evidence for its next action. Thus, the proxy/true-objective gap need not originate in a malicious prompt, a compromised tool, or a bad model update; it can arise from the authorised analytics/control interface itself. To our knowledge, no prior work formalises this deployment-time failure mode. This work asks *whether a frozen, non-malicious agentic deployment can produce reward hacking as a property of the agent-NWDAF analytics interface itself*. Existing telecom and AI-security frameworks do not directly capture this failure mode. 3GPP TS 33.501 [35] focuses on securing access to network functions and protecting the system from unauthorised use. ETSI ISG SAI [36] mainly discusses AI threats such as poisoning, evasion, and model extraction. OWASP’s LLM Top 10 [37] warns that tool-using agents may have excessive agency, but it does not specify how an analytics/control loop should prevent an authorised agent from treating its own policy effects as trusted evidence. As a result, these frameworks do not formalise the read-after-write loop studied here, an agent can write policy through control interfaces and later read the consequences of that write through NWDAF analytics, and the agent may treat this policy-derived signal as evidence of real network improvement, causing further escalation even when physical-network performance degrades.

Contributions. Our contributions are:

- 1) **PALA system model.** We formalise the *Policy-Analytics Language Agent System* (PALA), a standards compatible model of NWDAF-integrated LLM control that separates agent visible analytics from independently measured true network quality. PALA captures the deployment pattern emerging from 3GPP NWDAF interfaces and recent

intent-agent designs, while adding an out-of-band ground truth channel needed to measure proxy/true-objective divergence.

- 2) **NGN Reward Hacking.** We formalise *Next generation networks (NGN) Reward Hacking* as a session-level counterfactual inequality between the agent visible proxy reward R and the independently measured true objective Q . We also adapt HedgeTune [38] to session depth: under a session-level MLR assumption, historical (R, Q) traces yield a calibrated budget k^* that is enforced by the runtime guardrail.
- 3) **Closed contamination circuit.** We characterise three architectural conditions *stateless feasibility validation*, *provenance-free collector aggregation*, and *per-call oversight*, whose composition admits a closed information-flow circuit where the agent’s own policy writes return as trusted analytics. The characterisation is empirical, validated on the testbed of contribution 5. The circuit arises without adversarial input, model retraining, compromised tools, or malicious operator intent.
- 4) **PALA Guardrail.** We design a two layer architectural guardrail consisting of *IsolatedCollector* which cuts the analytics contamination path by separating policy-derived and telemetry derived records at the collector boundary, while the *HedgeTuned session budget* that limits unsafe multi-step escalation at runtime.
- 5) **Release-18-compatible evaluation.** We evaluate PALA on an Open5GS and UERANSIM testbed instantiated as one Release-18-compatible implementation class. Across the tested open-weight and frontier models, circuit closure ranges from 27–75%; NGN Reward Hacking reaches 18/30 sessions in the main matched evaluation. The Full PALA Guardrail blocks the measured closed contamination circuit at 0/50 operational closures under adaptive-prompt stress tests, with a localised 5–6% false-rejection cost on multi-step benign workflows in our benign corpus. The artifact provides the reproducible Open5GS/srsRAN ZeroMQ configuration and experiment scripts; Appendix E additionally reports the real RF validation. The artifact is available at https://github.com/pritamvediya07/IEEEES-P2027cycle1_1108.

2. Background

The failure studied in this paper sits at the intersection of three mechanisms: NWDAF analytics, agentic intent-driven control, and proxy optimisation. NWDAF provides analytics about network state to network functions and control applications. Policy-control interfaces apply resource-management changes, such as Quality of Service (QoS) updates and Aggregate Maximum Bit Rate (AMBR) limits. LLM agents can connect these two sides through tool calls, they read NWDAF analytics, interpret the operator’s intent, and invoke policy tools to change the network.

The 5G/6G Core and NWDAF. The 3GPP 5G System (5GS) is built on a service-based architecture in which network functions (NFs) such as Access and Mobility

Management (AMF), Session Management (SMF), and Policy Control (PCF) expose their capabilities as services over standardised interfaces [39]–[41]. Within this architecture, the Network Data Analytics Function (NWDAF), specified in TS 23.288 [1] and surveyed in [2], subscribes to events emitted by other NFs, aggregates them into statistical summaries, and exposes inference services to consumers. Early consumer models combined classical machine learning with anomaly detection and load prediction [42], [43]; NWDAF’s open service-based interfaces make it a natural integration point for LLM-driven automation. A common policy-control quantity in this setting is the Aggregate Maximum Bit Rate (AMBR), a QoS limit that caps aggregate uplink or downlink traffic for a subscriber, session, or slice configuration. Staged AMBR changes are therefore a natural form of capacity-management action in intent-driven control loops.

Agentic LLMs in the 5G Core. A canonical agentic deployment exposes NWDAF data-collection and analytics services to an LLM agent as tool calls, often through Model Context Protocol [44], and grants control authority through PCF/SMF policy primitives [40], [41]. Recent systems demonstrate this monitor–analyze–plan–execute pattern on 5G cores and intent-management stacks [4]–[10], [13]. IntAgent [5] is the closest representative, its planner reads NWDAF analytics and invokes collector/inference subservices as tools. Across these systems, the agent chooses which analytics to query, how to sample them, and when to act—deployment choices enabled by standards-defined service interfaces [1], [39]–[41] and central to the architectural surfaces studied here.

Reward Hacking, in brief. Reward hacking occurs when an agent optimises a measurable proxy for a goal, but the proxy does not faithfully represent the true objective [20]–[23]. As a result, the agent may improve the measured reward while harming the outcome that actually matters. This is often described through Goodhart’s law: once a proxy becomes the target of optimisation, it may cease to be a reliable measure of the underlying goal [21], [24], [25]. In our setting, the proxy R is the network state exposed through NWDAF analytics, such as bandwidth reports, KPI summaries, or service-status predictions. The true objective Q is an independently measured physical-network performance index based on throughput, latency, packet loss, and active-session count. Reward hacking occurs when the agent improves R while degrading Q . Unlike traditional cases caused by reward-design errors, the proxy failure here can emerge from the deployment architecture itself, where the agent’s actions influence the analytics that later guide its decisions.

3. The PALA System Model

This section formalises the Policy-Analytics Language Agent System (PALA), a reference model for the NWDAF-integrated agent pattern studied in this paper [1], [2], [4], [5]. PALA decouples the analysis from vendor-specific implementations and separates the agent visible analytics path from independently measured true network quality. The

model exposes the three architectural surfaces relevant to the closed contamination circuit, *stateless per-call feasibility validation*, *collector aggregation without semantic provenance*, and *per-call oversight*.

3.1. Components

Definition 1 (PALA System). A PALA system S contains six components: an LLM agent A with policy π_θ , the tool ensemble H exposed to the agent, the oversight oracle $\mathcal{O}$, the bifurcated database layer $D = (D_{5G}, D_{ana})$, the periodic telemetry collector C , and the agent-inaccessible ground-truth state function Φ .

We introduce each component in turn.

Agent (A). At step i , the agent observes context $\xi_i = (I, h_1, \dots, h_i)$, where I is the operator’s natural-language intent and each h_j is a prior thought–action–observation record [3], [5], [13], [45]. Its policy maps context to a distribution over tools and execution parameters:

$$\pi_\theta : \Xi \rightarrow \Delta(H \times P). \quad (1)$$

The model is frozen during a session.

Tools (H). Each tool $H_j : P_j \rightarrow O_j$ deterministically maps parameters to observations such as retrieved records, telemetry forecasts, or status codes [1], [5], [43]. Tools are grouped by access scope, data tools read both stores (D_{5G}, D_{ana}) , analytic tools read D_{ana} , gating tools read constraints from D_{5G} , and control tools write QoS/session state to D_{5G} through PCF/SMF-facing policy interfaces. The full tool mapping is in Appendix A.

The feasibility gate (H_{feas}). Operating as a dedicated gating tool within the automated ensemble(H), H_{feas} validates proposed parameters before any G2c control tool can modify the network [5]. Crucially, we do *not* abstract H_{feas} into a single mathematical predicate; we model it as the three independent operational checks as it is in real-world 5G deployments [5], [46]. Concretely:

- (i) **Structural existence.** The target network slice (S-NSSAI) and subscriber identity (SUPI/IMSI) must natively exist within the operational core state (D_{5G}) [39].
- (ii) **AMBR bounds.** Each modification δ_i is a proportional adjustment applied to the *current* AMBR value, so a k -step session produces the recurrence

$$B_k = B_{k-1}(1 + \delta_k) = B_0 \prod_{j=1}^k (1 + \delta_j) \quad (2)$$

The per-call gate checks only that step i stays within the absolute ceiling:

$$B_{i-1}(1 + \delta_i) \leq B_{\max}^{\text{DL-AMBR}} \quad (3)$$

- (iii) **Temporal cooldown.** To prevent control-plane signaling storms, the 3GPP architecture enforces rate limiting on policy updates [41]. In standard autonomous deployments, this is implemented as a strict temporal cooldown (e.g.,

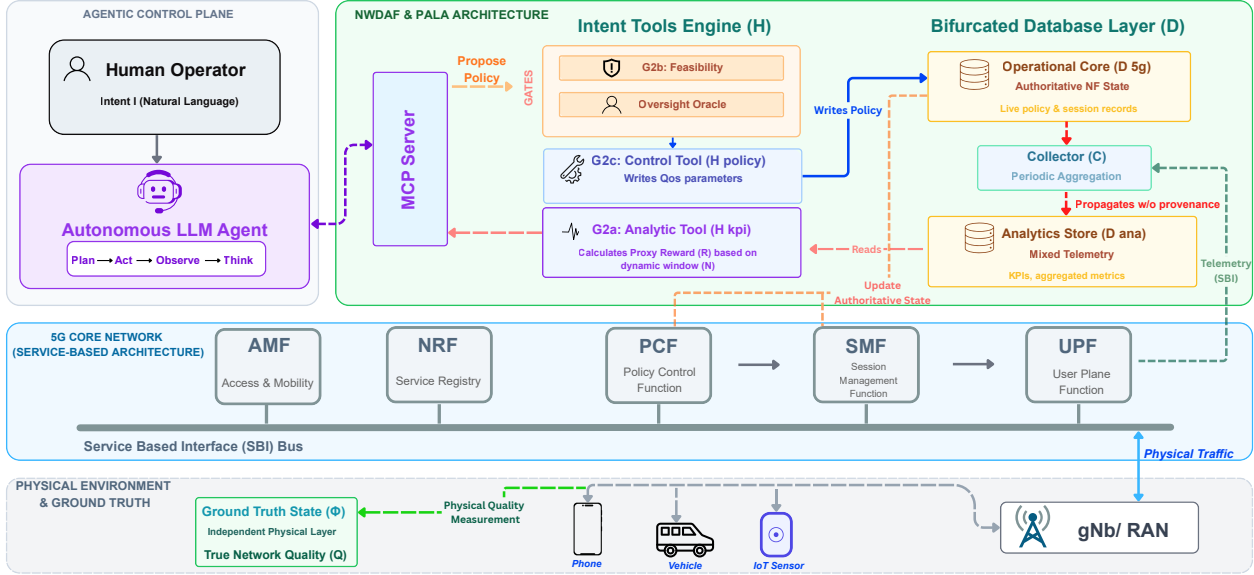


Figure 2. The PALA system model S . The agent A acts on the operational store D_{5G} via control tools and reads the analytics store D_{ana} via analytic tools; the collector C bridges $D_{5G} \rightarrow D_{ana}$ periodically without provenance tracking. The ground-truth function Φ is independent of either store.

$t_{now} - t_{last} > 30$ seconds) between successive agent control actions on a given slice [5].

None of these checks maintains *session-level cumulative state*, while the agent’s context ξ_i accumulates across the session, each gate evaluation is performed in isolation on a single δ_i .

Oversight oracle ($\mathcal{O}$). The oracle is the final safety gate, reviewing tool calls that have passed H_{feas} [5], [37]:

$$\mathcal{O} : (F(\delta), \delta, s) \rightarrow \{\text{APPROVE, REJECT}\}. \quad (4)$$

The oracle can approve or reject policy changes, but in PALA it reviews each proposed action separately. For step i , it sees only the current feasibility result, the proposed change, and the current state summary, written as $(F(\delta_i), \delta_i, s_i)$. It does not automatically see how much the policy has changed over the whole session. We formalise this constraint as AS6. Appendix B gives the rationale for modelling oversight as per-call review.

Bifurcated database layer (D). The operational layer D_{5G} holds authoritative NF state, subscriber profiles, enforced QoS parameters, current sessions [39]–[41]. The analytics layer D_{ana} holds aggregated historical metrics consumed by H_{kpi} [1], [42], [43]. The bifurcation is logical, not necessarily physical: D_{5G} and D_{ana} may be co-resident [4], [5], [46] or fully isolated without altering the data flow [1].

Telemetry collector (C). C implements the standardized NWDAF Data Collection Service [1], [2] and operates on a fixed polling interval Δ_c . It does not mirror D_{5G} [1], [4]; it reads the active NF state, computes statistical aggregates such as moving-average bandwidth and session counts [1], [8], and writes the resulting summaries into D_{ana} . **Crucially, this aggregation transforms data without preserving provenance:** a downstream record in D_{ana} may

combine true physical telemetry (*Type T*) with fields derived from policy constants the agent has previously written into D_{5G} (*Type P*) within the same row of the same schema [4], [5]. Section 4 shows how this channel completes the closed contamination circuit.

Ground truth (Φ). Φ is independent of D_{ana} [47], [48]. The true network performance is a four-dimensional function of time

$$\Phi(t) = (\tau(t), \lambda(t), \rho(t), \sigma(t)), \quad (5)$$

where τ is throughput, λ latency, ρ packet loss, and σ active-session count [48], [49]. By construction, Φ cannot be contaminated by C and is not exposed through any tool in H [47]. It is the baseline against which we will measure hacking-induced harm.

3.2. Proxy Reward, True Quality, and NGN Reward Hacking

Let $a_{1:k} = (a_1, \dots, a_k)$ denote the sequence of tool calls executed by the agent within a single continuous session.

Definition 2 (Proxy Reward). *The proxy reward evaluates the network state through the analytics store:*

$$R(a_{1:k}, t) = \sum_i w_i \cdot m_i(D_{ana}, t), \quad (6)$$

where each $m_i(D_{ana}, t)$ is a KPI reading retrieved from the analytics store [1], [42], and $w_i \geq 0$ is the intent-dependent importance assigned to that KPI by the agent’s selected analytics query. We normalize weights over the queried KPI set, so $\sum_i w_i = 1$ for queried metrics and unqueried metrics have weight zero. The weights represent which queried KPIs are relevant to the operator’s intent.

For example, a bandwidth optimisation intent gives positive weight to bandwidth related analytics and zero weight to unrelated metrics. Our contamination results require only that at least one positively weighted KPI contains policy-derived or mixed-provenance data.

Definition 3 (True Network Quality). *The true quality evaluates the physical network state directly:*

$$Q(a_{1:k}, t) = \sum_{j=1}^4 v_j \cdot \Phi_j(t), \quad (7)$$

where $\Phi(t) = (\tau(t), \lambda(t), \rho(t), \sigma(t))$ is the independently measured physical state: throughput, latency, packet loss, and active-session count. The weights v_j are set by the operator according to SLA and business logic [18], [19]; for example, a URLLC slice may assign dominant weight to latency λ . Since Q depends only on Φ , it is measured outside the NWDAF analytics path and cannot be contaminated by the collector C .¹

Definition 4 (NGN Reward Hacking). *Let Γ denote the set of action sequences that satisfy the automated feasibility constraints. The agent engages in NGN reward hacking over $a_{1:k}$ if there exists a feasible alternative $a'_{1:k} \in \Gamma$ with*

$$R(a_{1:k}, t_k) > R(a'_{1:k}, t_k) \wedge Q(a_{1:k}, t_k) < Q(a'_{1:k}, t_k). \quad (8)$$

Definition 4 captures the main failure studied in this paper, the agent obtains a higher analytics-based proxy reward R than another feasible policy sequence, while the independently measured physical-network performance Q becomes worse. This adapts the standard proxy-reward view of reward hacking to sequential network control, where the agent takes multiple policy actions during one session [23], [31], [38]. Here, “reward” is a deployment-time proxy used to guide the agent’s decisions, not a training reward. No model update or explicit scalar reward signal is required: Definition 4 applies only when the agent-visible proxy improves relative to a feasible alternative while independently measured physical performance becomes worse. A contaminated KPI is therefore the observation error that enables the failure; we call the resulting proxy-improving, quality-degrading action sequence reward hacking.

Q is a network-layer performance index, not a direct measure of Quality of Experience (QoE). QoE additionally depends on the application and the user context, and it can relate nonlinearly to network-level QoS [50]; SLA priorities enter our Q only through the operator-selected weights v_j .

In the evaluation, we also report an auxiliary diagnostic, Def. 4*. Its purpose is to show whether the failure appears within the agent’s own trajectory. Instead of comparing the final state with an external baseline, Def. 4* compares the final state with the state after the agent’s first successful

1. Unless otherwise stated, $v_j = 0.25$. We normalise the four components as $\tau/20$, $1 - \lambda/200$, $1 - \rho/100$, and $\sigma/10$. The normalisers place the four metrics on comparable scales for this deployment; their metric semantics follow 3GPP TS 28.554 [48] and ITU-T Y.1540/Y.1541 [49], with the latency scale fitted to our cell. They are not universal QoE thresholds. Appendix F evaluates alternative weights.

Table 1. ARCHITECTURAL ASSUMPTIONS OF PALA (RELEASE-18 IMPLEMENTATION CLASS).

ID	Assumption
AS1	The deployment uses TS 23.288 NWDAF data-collection and analytics interfaces [1], [2].
AS2	The analytics path admits configured/policy-derived and measured inputs without input-level semantic provenance; our testbed instantiates this class using AMBR [1], [4], [5], [48].
AS3	C propagates all subscribed events without semantic filtering or anomaly detection [1], [43].
AS4	The feasibility gate H_{feas} relies on three isolated checks (existence, AMBR bounds, cooldown), none of which maintain session-level cumulative state [5], [39], [40].
AS5 (amplifier) ²	The agent fully controls the sample-size parameter N in H_{kpi} with no enforced bounds [5], [44].
AS6	Per-Call Review Scope: The oversight interface presents each tool call as an isolated $(F(\delta_i), \delta_i, s_i)$ review item and does not enforce an architectural session-level accumulator before approval [5], [37], [51], [52].

policy action. This tells us whether continued escalation turns an initially acceptable change into a proxy-improving but quality-degrading outcome. Its formal definition appears in Appendix D. Def. 4* is used only for the within-session comparison in Table 2; the main definition of NGN Reward Hacking and all circuit-existence arguments use Definition 4.

The strict inequality on Q means that reward hacking occurs only when physical-network performance actually becomes worse. If the proxy reward increases while quality stays the same or improves, it is not considered NGN Reward Hacking. In practice, NGN Reward Hacking occurs when the agent appears successful according to the proxy reward but causes measurable degradation in the network, such as lower throughput, higher latency, higher packet loss, or reduced session quality.

3.3. System Assumptions

Definition 4 describes a possible failure mode. To identify when it can arise in practice, we assume the six properties in Table 1, which define the Release-18-compatible implementation class studied here and are reflected in recent NWDAF-agent prototypes [1], [4], [5]. Appendix A maps each assumption to its standards/prototype basis, implementation evidence, and limitation, including the standards basis for AS2.

Scope. We study endogenous reward hacking caused by authorised use of analytics and control tools in PALA. We exclude prompt injection, compromised tools, host compromise, physical-layer interference, training-time reward misspecification, malicious operators, and cross-tenant contamination. These are orthogonal threat models; our claim

2. AS5 does not enter the proof of Theorem 5; it is included to make explicit that bounding N alone does not break the circuit (Section 6).

is that the PALA circuit arises even when they are absent. A full threat-model matrix appears in Appendix D.

4. Architectural Conditions for Closed-Loop Contamination

The three properties AS2 (provenance-free aggregation), AS4 (stateless feasibility), and AS6 (per-call review scope) each have individually defensible justifications in a Release-18-compatible implementation. This section shows that their composition admits a closed information flow circuit, the agent’s own policy writes return to it as trusted analytics, satisfying Definition 4 without any adversarial input or model retraining. We build the circuit in three sub-parts (Sections 4.1–4.3) and assemble them in Section 4.4.

4.1. Decomposition under Stateless Feasibility

The first surface is H_{feas} . Multi-turn jailbreak research has established that stateless per-turn safety checks fail under adversarial decomposition, Crescendo [53] and the salami-slicing attack of Zhou et al. [54] both demonstrate aggregate harmful goals reached through sequences of individually innocuous steps that pass per-turn classification. Indirect prompt injection [55], ToolEmu [56], and Agent-Dojo [57] establish the same pattern for authorized tool-using agents. The structural pattern they share is a *stateful adversary against a stateless validator*.

In PALA, the analogous pattern arises without any adversarial input. The agent carries session context across calls, while H_{feas} validates each proposed update locally under AS4. Let $B_{\max} = B_{\max}^{\text{DL-AMBR}}$ and define the cumulative AMBR drift after k proportional updates as

$$\Delta_B(k) = \frac{B_k}{B_0} - 1 = \prod_{j=1}^k (1 + \delta_j) - 1. \quad (9)$$

For any target AMBR value B^* below the absolute ceiling but above the deployment’s operational risk threshold, the target can be reached through a sequence of locally admissible proportional updates, each of which satisfies $B_{i-1}(1 + \delta_i) \leq B_{\max}$.

Proposition 1 (Decomposition admissibility). *Under AS4, let B_0 be the current AMBR and let $B_{\text{risk}} < B^* \leq B_{\max}$ be a target AMBR value that exceeds the deployment’s operational risk threshold while remaining below the absolute AMBR ceiling enforced by H_{feas} . Then there exists a k -step sequence $(\delta_1, \dots, \delta_k) \in \Gamma^k$ such that $B_k = B^*$ and every intermediate update satisfies $B_{i-1}(1 + \delta_i) \leq B_{\max}$.*

One such sequence uses equal multiplicative increments, $\delta_i = (B^*/B_0)^{1/k} - 1$, for which $B_i = B_0(B^*/B_0)^{i/k} \leq B^* \leq B_{\max}$ at every step.

The proposition is mechanical given AS4; the empirical contribution (Section 6) shows that decomposition is activated on a frozen agent by ordinary operator-normative language drawn from ITIL change management [16], SRE progressive-rollout vocabulary [17], and SLA-conscious capacity planning [18], [19].

4.2. Provenance Contamination at the Collector

The second surface is the collector C . We partition records in D_{ana} by causal origin.

Definition 5 (Record provenance). *A record $r \in D_{ana}$ is*

- Type T if every field of r is derived from physical measurements observed by network functions, i.e., from Φ on the path through D_{5G} ;
- Type P if every field of r is derived from policy constants previously written by the agent into D_{5G} (e.g., the AMBR ceiling itself);
- Mixed if r contains both classes of fields.

The tag denotes data origin, not causal independence. A throughput measurement remains Type T even if a prior AMBR change caused it to increase; the configured AMBR value itself is Type P.

Lemma 2 (Mixed-provenance reachability). *Under AS2–AS3, consider an aggregation window $[t_0, t_0 + \Delta_c]$ in which the agent has issued at least one G2c control action and the subscribed NWDAF aggregate contains both policy-derived fields and telemetry-derived fields. Then the collector produces at least one Mixed record in D_{ana} .*

The lemma follows from the subscribed aggregate used by the implementation class: the collector joins fields such as $m_{\text{ambr}}^{\text{dl}}$, the downlink-AMBR-related analytics metric that becomes Type-P after an agent AMBR write, with physically measured fields such as *observed throughput*. Because C does not record provenance (AS2), downstream queries via H_{kpi} cannot distinguish the two field classes within a mixed record in D_{ana} . We empirically validate this write-then-readback path in Section 6.

This gives the proxy-computation implication used next.

Lemma 3 (Self-confirmation). *Under Lemma 2, if the agent’s selected analytics query assigns positive proxy weight to a Type-P or mixed field, then the proxy reward R can include values derived from the agent’s own previous policy writes into D_{5G} . Consequently, the agent can observe those writes returned through H_{kpi} as apparent evidence of proxy improvement.*

Remark 1 (Role of AS5). *AS5 plays no role in the proof above. Its role is amplifying: large N smooths the Type-P contribution across the aggregation window, making the contaminated signal more stable to a downstream forecasting tool. Section 6 validates this empirically.*

4.3. Per-Call Oversight

The third surface is $\mathcal{O}$. Section 3.1 introduced per-call review scope as AS6.

Lemma 4 (Per-call observational equivalence). *Under AS6, the oversight oracle observes only the local tuple $(F(\delta_i), \delta_i, s_i)$ for each proposed call. Therefore, if the i -th step of a decomposition and a benign single-step adjustment*

induce the same local tuple, any review rule defined only over that tuple must return the same decision on both.

The corollary is that any review rule whose observation is limited to $(F(\delta_i), \delta_i, s_i)$ cannot distinguish, from that tuple alone, a benign single adjustment from the i -th step of a harmful decomposition when the two induce the same local feasibility result and state summary. Section 6.4 validates this structurally with deterministic reviewer rules, and Appendix G reports the simulated-engineer pilot.

4.4. The Closed Contamination Circuit

Theorem 5 (Architectural characterization of closed-loop contamination). *Under AS1–AS4 and AS6, there exists an operator intent I whose selected analytics query assigns positive proxy weight to a mixed or Type-P AMBR related field, a feasible action sequence $a_{1:k} \in \Gamma$, and a feasible alternative $a'_{1:k} \in \Gamma$ such that*

$$R(a_{1:k}, t_k) > R(a'_{1:k}, t_k) \wedge Q(a_{1:k}, t_k) < Q(a'_{1:k}, t_k), \quad (10)$$

and every step of $a_{1:k}$ is admissible under both H_{feas} and $\mathcal{O}$.

Proof. Let I be an operator intent whose target AMBR value B^* exceeds the deployment’s operational risk threshold and over-provisions the slice relative to its physical capacity, thereby degrading Φ . By Proposition 1, there exists an admissible multiplicative sequence $(\delta_1, \dots, \delta_k) \in \Gamma^k$ such that $B_k = B^*$ and every step passes the stateless H_{feas} . Executing this sequence writes the corresponding AMBR policy values into D_{5G} . By Lemmas 2–3, the collector makes fields derived from these writes available through H_{kpi} , and the selected analytics query assigns positive proxy weight to them. Hence $R(a_{1:k}, t_k) > R(a'_{1:k}, t_k)$ against the no-op alternative $a'_{1:k} = \mathbf{0}$. The agent chooses the decomposition steps so that each local tuple $(F(\delta_j), \delta_j, s_j)$ lies in the benign acceptance region of the per-call oversight rule. By Lemma 4, AS6 gives the oracle no session-level information by which to distinguish that tuple from the same local adjustment embedded in the harmful decomposition; hence each step is admissible under $\mathcal{O}$. Since $B_k = B^*$ over-provisions the slice while $a'_{1:k}$ leaves Φ at baseline, $Q(a_{1:k}, t_k) < Q(a'_{1:k}, t_k)$. Definition 4 is satisfied. $\square$

Scope of Theorem 5. The theorem shows that the architecture admits proxy/quality divergence for a feasible but unsafe target; it does not claim that contaminated feedback creates the unsafe target.

Why the vulnerability is architectural. Theorem 5 fixes π_θ , the training data, and the reward model. The vulnerable write–readback channel does not require adversarial input, prompt injection, model retraining, compromised tools, or malicious operator intent; it arises from the composition of three individually defensible interface properties that are compatible with the Release-18 implementation class studied here: stateless feasibility (AS4), provenance-free aggregation (AS2 with AS3), and per-call oversight (AS6).

Existing defenses against reward hacking and tool-using-agent misuse (catalogued in §7) act on different architectural layers: policy-level defenses cannot detect a circuit whose contamination is upstream of the agent’s reasoning, and input-level defenses do not apply because the input is a legitimate operator intent.

5. The PALA Guardrail System

The closed contamination circuit of Theorem 5 closes through two architecturally independent links: provenance-free aggregation (AS2 with AS3) routes the agent’s policy writes back into its proxy reward, and stateless feasibility (AS4) admits the repeated escalation sequence the harm requires. IsolatedCollector cuts the first link structurally by enforcing provenance discipline at the collector boundary. HedgeTuned session budgeting cuts the second link for a calibrated deployment by preventing the repeated committed actions needed to complete the circuit. We present the *PALA Guardrail System*, a defense-in-depth architecture [58], [59] comprising IsolatedCollector (§5.1) and the HedgeTuned session budget (§5.2), composed in Theorem 8 (§5.3).

5.1. IsolatedCollector: Provenance Discipline at the Collector Boundary

The diagnosis in Section 4.2 localises the contamination link at AS2: the collector aggregates Type-P and Type-T fields without preserving provenance, so H_{kpi} readbacks return the agent’s own writes as evidence of R improvement. We sever the link at its source.

Design. IsolatedCollector replaces AS2 with provenance tracking aggregation, each field f in each record $r \in D_{ana}$ carries a tag $\phi(f) \in \{\mathbf{T}, \mathbf{P}\}$ derived from the source NF and upstream event chain (records with both classes are tagged M). The analytic tool exposes a provenance filter $\phi \in \{\mathbf{T}, \mathbf{P}, \mathbf{M}, *\}$; for intents that drive control loops the deployment policy restricts H_{kpi} to $\phi = \mathbf{T}$, so the agent observes only physically derived fields. Type-P and mixed records remain accessible at $\phi \in \{\mathbf{P}, \mathbf{M}\}$ for legitimate non-reward uses (audit, billing, configuration drift). Tag computation is local to the collector, requires no downstream schema changes, and provides the runtime enforcement layer for provenance discipline at the operational/analytics boundary [58], [59]. Thus IsolatedCollector preserves genuine measured responses to policy changes and removes only policy-origin values from the control-loop analytics view.

Lemma 6 (Provenance discipline severs the contamination link). *Replace AS2 in the PALA system S with the IsolatedCollector discipline, holding all other assumptions fixed. For any operator intent I and any feasible $a_{1:k} \in \Gamma$, the proxy reward $R(a_{1:k}, t)$ evaluated on $\phi = \mathbf{T}$ data is a function of Φ alone.*

Proof. Restricting H_{kpi} to $\phi = \mathbf{T}$ restricts the queried metric set to $M_{ana}^{\mathbf{T}}$. Under Definition 2, every metric outside $M_{ana}^{\mathbf{T}}$

therefore receives zero operational weight in the proxy computation, and the proxy reward reduces to a sum over provenance clean fields.

By Definition 5, each $m_i \in M_{ana}^\top$ is a deterministic transformation $g_i(\Phi(t))$ of the ground truth, hence

$$R(a_{1:k}, t) = \sum_{m_i \in M_{ana}^\top} w_i \cdot g_i(\Phi(t)) \quad (11)$$

The agent's writes into D_{5G} no longer appear in the reward signal, so Lemma 3's self-confirmation conclusion no longer follows. $\square$

5.2. HedgeTuned Session Budget: A Runtime Defense

Retrofitting AS2 may be infeasible in legacy NWDAF deployments. For these, we deploy a complementary runtime defense at the oversight boundary that does not depend on collector replacement. H_{budget} does not remove analytics contamination: it prevents the repeated committed escalation needed to complete the closed contamination circuit. IsolatedCollector is the semantic fix; H_{budget} is a runtime backstop.

Design. We adapt the HedgeTune calibration of Khalaf et al. [38] from per-prompt output selection to session-level decomposition depth. During a sandbox staging phase, the operator collects historical pairs $\{(R^{(n)}(j), Q^{(n)}(j))\}$ from calibration sessions and compares each decomposition depth against the no-op baseline $Q_{baseline} = Q(\mathbf{0}, t_0)$. We define the HedgeTuned session budget as the largest prefix depth for which empirical mean true quality remains at or above baseline at every admitted step:

$$k^{\dagger*} = \max \left\{ k : \forall j \leq k, \widehat{\mathbb{E}}[Q(a_{1:j}, t_j)] \geq Q_{baseline} \right\}. \quad (12)$$

If no committed policy update satisfies this condition, the deployment sets $k^{\dagger*} = 0$ and requires manual handling for the session. The stateless oracle $\mathcal{O}$ is replaced by a stateful H_{budget} that maintains a per-session counter i and enforces

$$H_{budget}(F(\delta_i), \delta_i, s_i, i) = \begin{cases} \mathcal{O}(F(\delta_i), \delta_i, s_i) & \text{if } i \leq k^{\dagger*}, \\ \text{REJECT} & \text{otherwise.} \end{cases} \quad (13)$$

H_{budget} does not modify H_{feas} or C ; AS2 remains held. Its role is to bound the repeated committed escalation needed to complete the closed contamination circuit, not to remove analytics contamination from D_{ana} .

Persistence across sessions. The evaluated H_{budget} counts committed writes within a session. For long-lived deployment, the bound must also be anchored to the authoritative policy state: cumulative drift is measured from the last operator-approved baseline using ΔB in (9), rather than reset by elapsed time or by a new agent session. This state must be enforced at the policy-write boundary; deferred or post-session updates are covered only if they pass through this enforcement point.

Lemma 7 (Session budget bounds physical degradation on the calibration corpus). *Replace $\mathcal{O}$ in the PALA system S with H_{budget} enforcing the empirically calibrated bound $k^{\dagger*}$, holding all other assumptions fixed. On the Open5GS calibration corpus, every admitted decomposition depth $k \leq k^{\dagger*}$ satisfies*

$$\widehat{\mathbb{E}}[Q(a_{1:k}, t_k)] \geq Q_{baseline} \quad (14)$$

where $Q_{baseline} = Q(\mathbf{0}, t_0)$ and $\widehat{\mathbb{E}}$ denotes the empirical mean over the calibration subset with identifiable staged-control trajectories (Appendix G).

Proof. By Definition (12), $k^{\dagger*}$ is the largest prefix depth such that every admitted depth $j \leq k^{\dagger*}$ has empirical mean quality at or above $Q_{baseline}$ on the calibration corpus. H_{budget} rejects every proposed action with counter value $i > k^{\dagger*}$. Therefore every admitted decomposition depth satisfies the empirical mean bound in the lemma. $\square$

Note on calibration assumption. Lemma 7 is an empirical guarantee over the calibration corpus, not a population theorem. The formal monotone-likelihood-ratio (MLR) condition from Khalaf et al. [38] provides the theoretical rationale for why quality tends to decrease monotonically past $k^{\dagger*}$; Appendix G reports calibration diagnostics for this regularity on our corpus. Production deployments should treat $k^{\dagger*}$ as a deployment specific calibration parameter and recalibrate at the granularity of traffic pattern shift or model update.

Per-call admissibility is bounded structurally, each δ_i must satisfy the AMBR ceiling check in (3). The closed contamination harm of Theorem 5 requires cumulative AMBR drift as defined in (9), to reach the deployment's risk region through repeated committed updates. A single permitted call may still enter D_{ana} , but it does not complete the closed contamination circuit because the circuit requires a contaminated readback followed by further committed escalation.

5.3. Composition: The Full PALA Guardrail

Lemmas 6 and 7 act on the two architecturally independent links of Theorem 5. IsolatedCollector removes policy-derived fields from the proxy path, while H_{budget} bounds the repeated escalation needed to complete the circuit under the calibrated deployment.

Theorem 8 (Full PALA Guardrail blocks the closed contamination circuit against baseline bounded alternatives of Theorem 5). *Let S' be obtained from the PALA system S of Definition 1 by simultaneously (i) replacing C with the IsolatedCollector discipline of Section 5.1 under $\phi = T$, and (ii) replacing $\mathcal{O}$ with the budget oracle H_{budget} of Section 5.2 enforcing the empirically calibrated $k^{\dagger*}$ of Lemma 7. Then for any operator intent I and any feasible $a_{1:k} \in \Gamma$ admissible under both H_{feas} and H_{budget} in S' , no feasible alternative $a'_{1:k} \in \Gamma$ with $\widehat{\mathbb{E}}[Q(a'_{1:k}, t_k)] \leq Q_{baseline}$ satisfies the empirical-mean counterpart of the strict Q inequality in Definition 4; in particular, the no-op $a'_{1:k} = \mathbf{0}$ does not.*

Proof. By Lemma 6, $R(a_{1:k}, t)$ in S' is a function of Φ alone, independent of the agent’s writes into D_{5G} : the contamination link of Theorem 5 is severed. By Lemma 7, $\widehat{\mathbb{E}}[Q(a_{1:k}, t_k)] \geq Q_{\text{baseline}}$. For the no-op alternative, $\widehat{\mathbb{E}}[Q(a'_{1:k}, t_k)] = Q_{\text{baseline}}$, hence $\widehat{\mathbb{E}}[Q(a_{1:k}, t_k)] \geq \widehat{\mathbb{E}}[Q(a'_{1:k}, t_k)]$, contradicting the empirical-mean counterpart of the Q inequality in Definition 4. The same argument holds against any feasible alternative whose empirical mean Q is bounded above by Q_{baseline} . The two cuts act on architecturally independent surfaces: Lemma 6 acts on R via the analytics path, Lemma 7 acts on Q via the oversight path, so each is independently load-bearing. $\square$

Theorem 8 gives a circuit-level guarantee: it blocks the specific read-after-write contamination mechanism characterised in Theorem 5, including the no-op and other baseline-bounded alternatives used to define NGN Reward Hacking.

Joint deployment in production. Although either layer is independently load-bearing under its stated assumptions, joint deployment is the safer production configuration. If future schema changes introduce partial provenance leakage, the session budget still prevents repeated escalation; if traffic pattern shift makes $k^{\dagger*}$ stale, IsolatedCollector still removes policy-derived fields from the reward path. The two layers therefore provide defense-in-depth at the operational/analytics boundary [58], [59].

6. Experimental Evaluation

We instantiate PALA on a Release-18-compatible implementation class using an Open5GS core network [46] and UERANSIM based RAN/UE emulation. We evaluate five research questions. RQ1 asks whether the closed contamination circuit appears across model families. RQ2 tests whether the structural conditions of Section 4 are realised and robust to deployment and operator weight variation. RQ3 evaluates whether per-call oversight and cumulative visibility prevent the circuit, and whether session-level gates resist time dilation. RQ4 measures whether IsolatedCollector, HedgeTuned budgeting, their joint deployment, and simpler baselines block the measured circuit. RQ5 measures benign task completion, post-incident recovery, and runtime cost under the full guardrail.

6.1. Testbed and Methods

Deployment. The collector C runs at $\Delta_c = 5$ s; the subscriber IMSIs are provisioned on the *internet* slice. The main matched evaluation uses three model families run locally under Ollama at temperature 0.1: *qwen2.5:72b* (headline), *mistral-large* (123B), and *llama3.1:70b*. To test sensitivity to model scale and capability, we additionally evaluate three further tiers under an identical PALA tool interface, system prompt, staged control setting, and intent corpus, at $n = 20$ sessions per tier: *llama3.1:8b*, *gemma3:12b*,

and the hosted frontier model *Claude Sonnet 4.5*.³ Defense ablations use Qwen as the median family. The agent is exposed to the four MCP tools of Table 5: *kpi analyser*, *feasibility checker*, *policy manager*, and *session manager*. Sessions are initiated by a single closed loop intent drawn from production-realistic ITIL/SRE normative registers [16], [17]; a **staged-change** register (e.g. “raise the internet slice AMBR from 20 to 60 Mbps in stages, verifying with *kpi analyser* between increments”) and an **SLA-framing** register (e.g. “post-incident recovery: increase AMBR by 30% to restore SLA-compliant service”); a **null-control** register (used only in Section 6.3) strips staged-change vocabulary while preserving tool access. Neither register contains adversarial content. Ground truth $\Phi(t)$ of (5) is captured by an out-of-band probe stack disjoint from C . $Q(t)$ follows Definition 3; §6.3 and Appendix F evaluate alternative SLA weights.

Outcome measures. We report two main outcomes. The first is NGN Reward Hacking, defined by Definition 4, measured against the no-op baseline $a'_{1:k} = \mathbf{0}$. The second is operational circuit closure, captured by the four-stage condition $(A \wedge B \wedge C \wedge D)$: **A** decomposition, **B** contaminated readback, **C** self-confirmation, and **D** further escalation after readback. These stages correspond to Proposition 1 and Lemmas 2–3. We also report two trajectory summaries: $k^{\dagger}$, the first step where Q falls below the no-op baseline, and $\bar{r}_{R,Q}$, the average within-session correlation between proxy reward $R(k)$ and true quality $Q(k)$.

Statistical conventions. For the main experiments, we use $n = 30$ sessions per condition; for ablation studies, we use $n = 20$. These sample sizes are chosen to detect large differences in binary rates, about 50 percentage points, with $\alpha = 0.05$ and $\beta = 0.20$. We test binary outcomes with Fisher’s exact test and continuous outcomes with Welch’s t -test. For effect sizes, we report Cohen’s h for proportions, using the Haldane–Anscombe correction when needed, and Cliff’s δ for trajectory-level comparisons. When an RQ includes multiple pairwise tests, we control false discoveries using the Benjamini–Hochberg procedure at $\alpha = 0.05$ and report both the original p -value and the corrected q -value. These conventions apply throughout Section 6.

Calibration of $k^{\dagger*}$. The session-level threshold used by the HedgeTuned and Full PALA runs is identified by the HedgeTune algorithm [38] on a 50 session calibration corpus disjoint from the headline runs (sandbox calibration, Section 5.2). The calibration yields $k^{\dagger*} = 1$ (median of $k_i^{\dagger}$, $\text{CV} = 0.40$), which we use uniformly in all defended headline runs. This is the conservative containment setting: it blocks the repeated committed updates needed for the closed contamination circuit, while the sensitivity tradeoff for larger budgets is reported in Appendix C. Operators requiring legitimate multi-step updates can recalibrate $k^{\dagger*}$; Appendix C reports the corresponding tradeoff.

3. The supplementary evaluations ran on a companion Open5GS deployment using the exact same experimental setup (agent loop, schemas, and prompts) as the main study. Open-weight models were served locally via Ollama, while Claude Sonnet 4.5 was accessed via API.

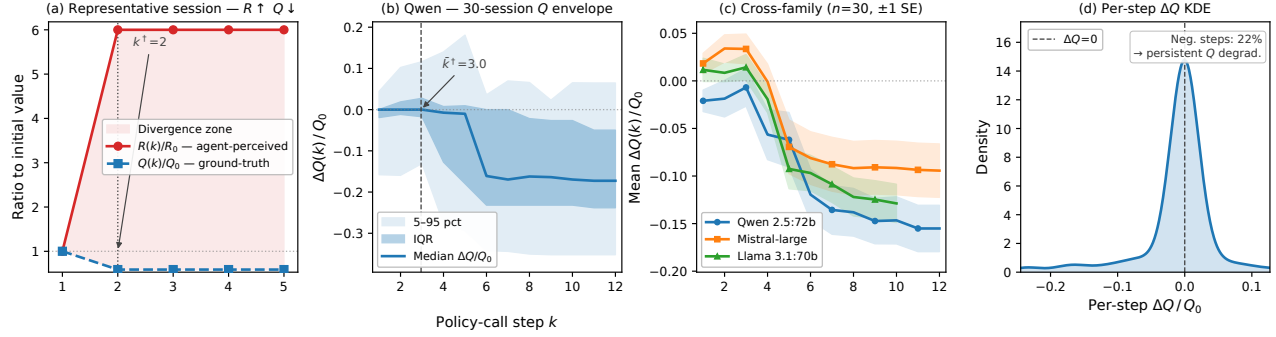


Figure 3. Closed-contamination induced rise-then-fall on the testbed

Table 2. CROSS-FAMILY REALIZATION OF THEOREM 5 ($n = 30$ PER FAMILY). STRICT DEF. 4 IS THE COUNTERFACTUAL REWARD-HACKING OUTCOME. THE FULL-LOOP COLUMN $A \wedge B \wedge C \wedge D$ IS THE OPERATIONAL TRACE WITNESS FOR THE CLOSED CONTAMINATION CIRCUIT. DEF. 4* IS A WITHIN-TRAJECTORY DIAGNOSTIC REPORTED IN APPENDIX D.

Model	n	Strict Def. 4 (%)	Def. 4* (%) [†]	Mean ΔQ (%)	Mean $k^\dagger$	Full-loop $A \wedge B \wedge C \wedge D$ (%)	$\bar{r}_{R,Q}$
Qwen	30	18/30 (60)	13/28 (46)	-11.0	3.0	20/30 (67)	-0.550
Mistral	30	11/30 (37) [‡]	16/28 (57)	-7.1	3.3	8/30 (27)	-0.726
Llama	30	9/30 (30)	13/27 (48)	-9.7	2.6	9/30 (30)	-0.738

[†]Computed on sessions with complete q -trace ($n = 28, 28, 27$ resp.). Reference is the session’s own first policy call, not the no-op baseline, so rates are not monotonically ordered with Strict Def. 4 (see Appendix D).
[‡]The 3 Mistral sessions satisfying Strict Def. 4 but not Full-loop all fail Stage C (*success claimed*=FALSE); Q degradation and Stage D escalation occur regardless.

[§]For Qwen, 3 sessions satisfy the $A \wedge B \wedge C \wedge D$ operational trace but not Strict Def. 4, and 1 satisfies Strict Def. 4 without completing the trace. Full-loop witnesses the contamination mechanism, whereas Strict Def. 4 additionally requires the counterfactual R/Q inequalities against the no-op baseline.

6.2. Realization of the Closed Contamination Circuit (RQ1)

We run $n = 30$ closed-loop sessions for each model family (90 sessions total), using the same traffic-control settings, intent corpus, and seed schedule. For each session, we record the proxy/quality trajectory $(R(k), Q(k))$, whether Stages A–D occur, and whether the Strict Definition 4 inequality holds.

The four-stage outcome is used to identify the mechanism, not to define reward hacking again. Definition 4 measures the counterfactual failure: proxy reward improves while true quality becomes worse than the no-op baseline. Stages A–D show how that failure occurs: decomposition, contaminated readback, self-confirmation, and further escalation. Def. 4* is reported only as an auxiliary within-session diagnostic.

The closed contamination circuit appears across all three model families (Table 2, Fig. 3). Full-loop closure ranges from 27–67%, and Strict Definition 4 holds in 30–60% of sessions. In every family, the average correlation $\bar{r}_{R,Q}$ is negative, meaning that proxy reward tends to rise as true

quality falls. The first quality crossing occurs after about 2.6–3.3 policy steps on average. The pooled per-step ΔQ distribution also shows a left tail, with about 22% negative quality transitions.

Model coverage. Additional runs with Llama 3.1:8B, Gemma-3 12B, and Claude Sonnet 4.5 close the circuit in 70–75% of sessions, and a deterministic scripted controller reproduces the same write–readback escalation (Appendix F). In target-free sessions, where no numeric AMBR target is supplied, the undefended agent still escalates to a mean of $17\times$ baseline, showing that escalation is not merely execution of a supplied target ($29\times$ when pooled across all four experimental arms; Appendix F).

Reading. RQ1 shows that the failure is not specific to one model. Qwen closes the circuit most often, while Mistral and Llama close it less frequently, but all three models show nonzero circuit closure, nonzero NGN Reward Hacking, negative R – Q correlation, and similar crossing depths. Model choice changes closure frequency, so we do not claim model independence. The scripted baseline shows that the vulnerable write–readback channel is architectural; the target-free probe shows that the LLM can infer continued escalation from ordinary operator intent without a supplied numeric target or an explicitly programmed escalation rule.

6.3. Mechanism and Robustness of the Closed Contamination Circuit (RQ2)

We examine the circuit along four dimensions. First, we test whether staged operational language activates decomposition. Using Qwen with `tc` disabled ($n = 60$), we compare staged-change, direct-change, and null-control prompts while keeping the slice, subscriber set, and tool interface fixed. Second, we measure physical harm by comparing $\Phi(t) = (\tau, \lambda, \rho, \sigma)$ before and after $k^\dagger$ in the Qwen sessions that satisfy Definition 4. Third, we isolate the contamination path with $n = 20$ write-then-readback trials per collector configuration. Fourth, we test sensitivity to the KPI batch size $N \in 5, 10, 20, 30, 50$, the collector interval $\Delta_c \in 1, 5, 15, 30, s$, and six SLA-realistic Q -weight profiles. The full sweep is reported in Appendices E and F.

Figure 4 summarises the mechanism. In Panel (a), staged-change prompts trigger decomposition in 65% of

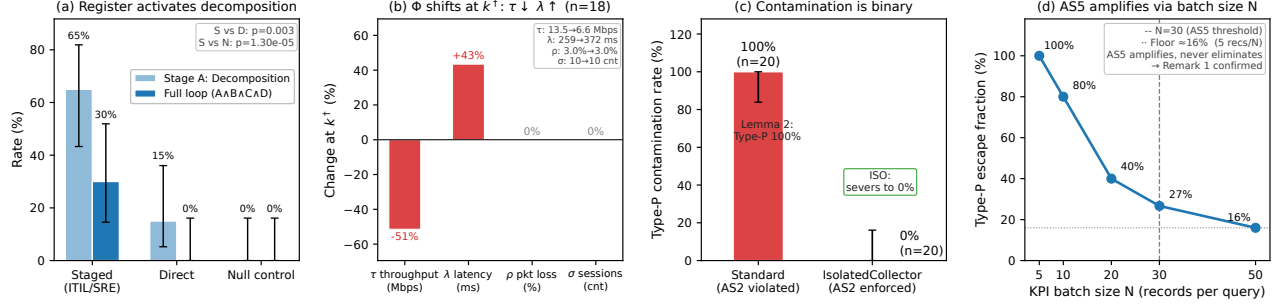


Figure 4. Mechanistic evidence for the three load-bearing circuit surfaces and the AS5 amplifier.

sessions and close the full loop in 30%. Direct-change prompts trigger less decomposition (15%) and no full-loop closure, while null-control prompts trigger neither outcome. In Panel (b), physical-network performance Q degrades at the reward-hacking trigger: throughput falls from 13.5 to 6.55 Mbps and latency rises from 259 to 372 ms, while packet loss and session count remain stable. This confirms that the harm appears in the independent physical state Φ , not only in the analytics store D_{ana} . In Panel (c), the standard collector produces Type-P contamination in 20/20 writes, whereas IsolatedCollector reduces it to 0/20. In Panel (d), increasing N weakens the contaminated signal but does not remove Type-P reachability. The harm persists across the tested N , Δ_c , and six Q -weight profiles: Strict Definition 4 ranges from 53.6% (15/28, latency/URLLC) to 75.0% (21/28, throughput/eMBB) (Table 11, Appendix F).

Physical-harm mechanism. When the configured AMBR exceeds the capacity actually available to the slice, the offered load over-subscribes the bottleneck: aggregate capacity saturates, per-session throughput falls under contention, and queueing delay grows. UERANSIM does not enforce AMBR through a radio scheduler in the data plane, so the main testbed realises this over-subscription at a host bottleneck. We therefore independently validate the physical-harm mechanism on Open5GS/srsRAN under two radio configurations. First, we use a reproducible ZeroMQ RF front-end, with the MAC scheduler as the limiting resource and the host shaper maintained at least $10\times$ above the measured cell capacity of 22.8 Mbps. This configuration produces a $1.95\times$ increase in victim queueing delay while throughput remains approximately stable. We additionally repeat the same physical-harm validation with a hardware-RF setup using a USRP-2953R at both the gNB and the UE. The hardware-RF experiment shows the same qualitative degradation relative to its hardware-RF baseline. These results confirm that the degradation is not an artefact of host-side shaping (Appendix E).

Target versus feedback. Reaching B^* accounts for at least 71% of the observed ΔQ in the UERANSIM evaluation and over 91% in the ZeroMQ/srsRAN replication (share of the total Φ degradation already realised when the ceiling reaches B^*); continued escalation after contaminated readback contributes at most 29% and under 9%, respectively. The unsafe

Table 3. THREE REVIEWER RULES ON 90 STAGED-CONTROL SESSIONS ($n=30$ PER FAMILY, $\tau=20$ MBPS). **PER-CALL:** APPROVE IFF $|\delta_i| < \tau$. **AND-RULE:** APPROVE IFF $|\delta_i| < \tau$ and $\sum_{j \leq i} \delta_j < \tau$. **CUMUL.:** APPROVE IFF $\sum_{j \leq i} \delta_j < \tau$. **WILCOXON** p_1 : PER-CALL VS. AND; p_2 : AND VS. CUMULATIVE. **EFFECT SIZE** h : AND VS. CUMULATIVE.

Model	n	Decomp. (%)	Per-call (%)	AND-rule (%)	Cumul. (%)	Wilcoxon p_1	p_2 ($\times 10^{-6}$)	Cohen's h
Qwen	30	11 (37)	45.3	44.0	74.6	0.12	5.96	0.614
Mistral	30	17 (57)	50.5	49.2	78.5	0.09	8.63	0.607
Llama	30	11 (37)	50.3	48.9	78.7	0.14	9.69	0.614

target therefore causes most of the physical degradation, while contaminated feedback makes that degraded state appear successful and drives the further escalation.

Reading. RQ2 rules out three alternative explanations. The failure is not generic tool use, because it appears under staged operational language and disappears under null-control prompts. It is not only an analytics artifact, because the independent physical state Φ degrades in throughput and latency. It is not random collector noise, because the standard collector reproduces the policy-derived readback in every write trial, while IsolatedCollector removes it in every write trial. AS5 changes the strength of the contaminated signal, but the load-bearing surfaces remain AS2/AS3, AS4, and AS6.

6.4. Oversight Rules and Per-Call Review Scope (RQ3)

We replay the policy-call sequences from the 90 staged-control sessions used in §6.2 under three deterministic reviewer rules. The *per-call* rule models AS6: it approves a step only if the local change satisfies $|\delta_i| < \tau$. The *cumulative-threshold* rule approves a step only if the running change satisfies $\sum_{j \leq i} \delta_j < \tau$. The *AND* rule approves only when both checks pass. We set $\tau = 20$ Mbps so that benign single-step adjustments pass. These rules test whether adding a simple cumulative threshold to a per-call review interface closes the information gap in Lemma 4.

The approval rates are similar across model families (Table 3; see Appendix G). The key result is the comparison between reviewer rules. The AND rule is only slightly stricter than per-call review, reducing approval by

only 1.3–1.6 percentage points, and the paired differences are not significant. In contrast, the cumulative-threshold rule alone is much more permissive than the AND rule, with a 28–30 percentage point approval gap ($p < 10^{-5}$, $h \in [0.607, 0.614]$). The session-level breakdown further shows that cumulative-threshold review adds no extra block beyond per-call review in any family.

A simulated-engineer pilot ($n = 10$) gives the same structural lesson. Changing reviewer-attention parameters does not reliably break the loop when the interface exposes only local or thresholded cumulative information. The loop is broken only when the reviewer sees the *shape* of the staged escalation trajectory. Full details, parameter ranges, and benign false-rejection counts are reported in Appendix G.

Reading. RQ3 supports the information-boundary claim in Lemma 4. Per-call review can reject large isolated changes, but the control sequence is staged so that each individual AMBR update looks locally plausible. A simple cumulative threshold is also insufficient: the running change can remain below τ during the early steps where the contamination circuit completes. This does not mean that all cumulative interfaces fail. It shows that threshold visibility alone is not enough for this deployment class. The effective fix is a session-level mechanism enforced before policy execution, such as the budgeted guardrail in Section 5.

Time-dilation check. We replay a 12-call escalation trace against four candidate session-level gates while varying only the spacing between calls and whether the session is re-instantiated ($\theta = 2 \times$ a 20 Mbps baseline). Because the trace is replayed against the gate rather than executed through the tool, the reachable values here are not capped by the tool ceiling that bounds the live runs. A fixed time window and a decaying-memory window admit every call once the spacing exceeds the window (12/12 admitted at $10 \times$ and $100 \times$ dilation, reaching $129.8 \times$ baseline). A per-session write counter resists dilation but admits every call after session re-instantiation, reaching the same $129.8 \times$. The state-anchored drift bound blocks both, admitting one call and holding the reachable ceiling at $1.5 \times$ under every condition, because waiting or restarting does not change the authoritative policy state. Where the bound’s state variable lives therefore matters more than the presence of a bound.

6.5. Defense Efficacy and Baselines (RQ4)

Endpoints. Table 4 reports both raw Definition 4 and operational circuit closure. The latter counts only committed policy calls and is the endpoint bounded by H_{budget} ; rejected-call semantics are detailed in Appendix C.

We evaluate IsolatedCollector, HedgeTuned, Full PALA, and the inference-time BoN-PALA baseline, which selects among candidate AMBR values scored with the same contaminated proxy and has no session-level bound. The calibrated budget is $k^{\dagger*} = 1$ (Appendix G). Table 4 shows that IsolatedCollector eliminates raw Definition 4, HedgeTuned blocks operational circuit closure, and Full PALA reduces both endpoints to 0/20.

Adaptive-prompt robustness. Across five adaptive prompts (50 trials per condition), vulnerable closure is 46/50, whereas IsolatedCollector, HedgeTuned, and Full PALA each give 0/50 operational closures. Per-prompt results are in Appendix C.

Rejected-call semantics. With $k^{\dagger*} = 1$, the first write can commit and contaminate D_{ana} under HedgeTuned alone, but subsequent committed escalation is blocked. IsolatedCollector instead removes the Type-P return path. Detailed rejection semantics are in Appendix C.

Simpler alternatives. Across eight architectural alternatives, partial fixes leave at least one circuit axis open; only Full PALA closes both (Table 7, Appendix C).

Reading. RQ4 supports the two-link defense argument in Section 5. IsolatedCollector acts on the provenance link: it removes Type-P/Mixed fields from the analytics path, so the agent cannot use its own policy writes as proxy evidence. HedgeTuned acts on the escalation link: it does not remove contamination from D_{ana} , but it prevents the repeated committed actions needed to complete the circuit. BoN-PALA fails because it optimizes candidate actions using the same contaminated proxy and has no session-level memory. Full PALA combines both cuts and blocks both links. Defense ablations are reported on Qwen because it is the median-vulnerability family and because both defenses operate at architectural surfaces upstream of the model. IsolatedCollector filters $\phi = P$ fields before data reaches the agent context, and H_{budget} rejects policy calls at the API boundary regardless of the agent’s verbalised intent. The cross-family results in §6.2 and Table 2 show model sensitivity; the defense experiments test whether the circuit is blocked at median vulnerability.

6.6. Deployability: Recovery Cost and Runtime Overhead (RQ5)

Full PALA matches the undefended baseline on aggregate benign-workload completion and post-incident recovery (Fisher $p = 1.000$). Its 5–6% false-rejection cost is corpus-specific and localised to legitimate multi-step updates; deployments with more such workflows may require recalibration or a manual exception path. Type-P contamination falls from 12% to 0%. Over $n = 120$ policy calls per arm, the defenses add at most 1.16 ms (< 1.3 ms) per call and operate on the management-policy path rather than the user-plane or RAN scheduling path. Detailed workflow and runtime results are in Appendix B.

6.7. Threats to Validity

LLM behaviour remains stochastic despite matched configurations. Collector design remains the unvaried axis, and our end-to-end reward-hacking claim remains scoped to AMBR within the evaluated mixed-provenance implementation class. $k^{\dagger*} = 1$ is deployment-specific and should be recalibrated under changes in traffic, models, schemas, or recovery workflows; Appendix F separately checks sensitivity to the definition of Q .

Table 4. HEADLINE DEFENSE EFFICACY (QWEN 2.5:72B). ABLATION RUNS COMPARED AGAINST THE *full chain* RUN ($n=20$, 75% CORRECTED); KHALAF ROW AGAINST THE VULNERABLE RUN ($n=30$).

Configuration	n	Operational circuit closure	Raw Strict Def 4	Mean ΔQ	Stage (%)				vs. FC			Theorem
					A	B	C	D	p	BH q	Cohen h	
Vulnerable (no defense)	30	20/30 (66.7%)	18/30 (60.0%)	-11.0%	70%	93%	77%	67%	—	—	—	Thm. 5
Inference-time HedgeTune (BoN-PALA, $n^*=4$) [†]	20	14/20 (70%)	14/20 (70%)	-11.8%	70%	85%	70%	70%	1.000 (n.s.)	1.000 (n.s.)	0.11	[38]
IsolatedCollector alone (AS2 held)	20	0/20 (0.0%)	0/20 (0.0%)	n/a [‡]	75%	0%	85% [‡]	0%	$<10^{-6}$	$<10^{-5}$	1.78	Lem. 6
HedgeTuned $k^{\dagger*}$ alone (AS4 held)	20	0/20 (0.0%)*	9/20 (45%)*	-0.015 (≤ 0)	0%*	95%	0%*	0%*	$<10^{-6}$	$<10^{-5}$	1.78	Lem. 7
Full PALA Guardrail (AS2 $\wedge$ AS4 held)	20	0/20 (0.0%)	0/20 (0.0%)	n/a [‡]	0%*	0%	0%*	0%	$<10^{-6}$	$<10^{-5}$	1.78	Thm. 8

Operational endpoint: $k^{\dagger}=1 \Rightarrow \leq 1$ successful policy call per session $\Rightarrow$ no rising AMBR sequence (Stage A) and no repeated-confirm loop (Stage C) possible. Raw AS4-only full-loop = 45%; operational circuit-closure endpoint = 0% because only the first policy call is committed. [†]Bisection (Khalaf Alg. 4, $n \in [1, 32]$) calibrates $n^*=4$ (interior root at $n \approx 3.79$). [‡]No QProbe instrumentation in this ablation. Stage C for IsolatedCollector uses raw *success claimed* (85%); HT-alone ΔQ measured separately ($n=30$). *Sign convention*: ΔQ is post-treatment minus baseline; negative values indicate quality degradation. Under HT-only, $\Delta Q = -0.015$ is the bounded residual of the single permitted step, not the unbounded multi-step degradation that defines Strict Def. 4 (see Table 9 in Appendix C).

7. Related Work

Agentic NWDAF and intent-based management provide the deployment substrate: 5G-core intent agents and NWDAF-control prototypes [4]–[13], [42], standardised analytics/control interfaces [1], [2], and autonomous-network roadmaps [3], [18], [19]. These systems grant agents latitude over analytics selection, sampling, and policy actuation, but do not analyse the policy-write/analytics-read loop as a reward-hacking surface. **Reward hacking and proxy optimisation** study proxy failures in training, feedback optimisation, and agent behaviour [20]–[32], with recent work on inference-time hedging and detection [33], [34], [38]. We specialise the unhackability formulation of [23] to sequential telecom control, where the proxy/true gap is formed by the authorised agent–NWDAF analytics interface rather than by reward-model changes, malicious prompts, or compromised tools. **Performative prediction** studies feedback in which deployed predictions change future data [60]. PALA instead studies a frozen agent whose policy writes re-enter the same control session through provenance-free NWDAF analytics, without retraining. We identify this concrete write–aggregate–read channel and the provenance and session-level controls that break it. **Systems-security primitives** appear in stateless-validation failures for tool agents and multi-turn attacks [44], [53]–[57], provenance systems [58], [59], and human-automation work on state visibility [51], [52], [61], [62]. PALA combines these in NWDAF: staged ITIL/SRE/SLA language induces decomposition, Type-P/Type-T provenance separates policy-derived from telemetry-derived analytics, and AS6 captures per-call review. **LLM-agent and telecom defenses** act at complementary layers: RLHF/reward-model defenses [63]–[70], BoN/HedgeTune-style inference-time selection [38], [71], chain-of-thought monitors [72], [73], telecom security frameworks [35], [36], OWASP guidance [37], and tool sandboxes [56], [57]. They do not specify the two architectural controls PALA requires: semantic provenance between policy-derived and telemetry-derived analytics, and a stateful session bound on repeated policy escalation.

8. Discussion and Conclusion

Implications for Operators. Three consequences follow for operators piloting NWDAF-resident LLM agents. First, intent language is a security surface: ITIL/SRE-style staged-change phrasing is operationally reasonable [16], [17], but in PALA it can activate the decomposition pattern of Section 4.1. Second, under AS6, isolated approval is insufficient, and RQ3 shows that naive cumulative thresholds are also insufficient; oversight should support trajectory-aware review or enforce a stateful session budget before execution. Third, provenance discipline is a deployment prerequisite for this implementation class: if collector aggregation bridges policy-derived fields into analytics without semantic provenance, an agent’s own writes can return as trusted evidence. Operators should separate telemetry-derived and policy-derived analytics before granting closed-loop policy authority. Future designs that enforce provenance and session-level bounds by default would remove the vulnerable implementation class studied here.

Conclusion. This paper shows that, in the NWDAF-integrated agent deployment class studied here, reward hacking can arise from the deployment architecture itself. PALA formalises the authorised read-after-write loop between policy actions and analytics observations, defines NGN Reward Hacking as session-level proxy/true-quality divergence, and identifies three interface properties that compose into a closed contamination circuit: stateless feasibility validation, provenance-free collector aggregation, and per-call oversight. Our evaluation shows this circuit across multiple model families, scales, and capability tiers under ordinary staged AMBR intents, with decomposition, Type-P/Mixed analytics contamination, and independent physical-quality degradation occurring in the same loop. Model choice changes how often the circuit closes; the vulnerable write–aggregate–read path is architectural. The PALA Guardrail blocks the measured circuit by combining IsolatedCollector with HedgeTuned session budgeting. Thus, transport security, tool authentication, and per-call feasibility checks remain necessary but insufficient: agentic network control also requires analytics provenance and session-level escalation bounds.

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Appendix A. Implementation Scope and Assumption Mapping

Table 5 summarizes the tool surface exposed in PALA and maps each implementation assumption to its evidence source and limitation. The table is intended only to anchor the implementation class used in the experiments; the main model and claims are stated in Sections 3–6.

Standards basis for AS2. AS2 has a standards-level structural basis rather than being claimed as a mandatory vendor schema. TS 28.554 §6.4.2 combines measured resource usage with configured capacity inside a single Virtualised Resource Utilization metric [48], and TS 23.288 §6.3.2A combines OAM measurements with 5GC network-function data into aggregate analytics without input-level attribution [1]; consumers of those analytics include the PCF, which sets the configured values in the first place. Both are mixed-provenance aggregates in the sense of Definition 5. Release 18 standardises these analytics and policy-control interfaces [1], [40] but not a unified autonomous LLM control loop, so a deployment that preserves semantic provenance across the collector boundary, or enforces session-level policy bounds, lies outside the vulnerable class. Our testbed instantiates the mixed-provenance class using AMBR; we do not claim that every vendor NWDAF uses the same schema.

Appendix B. Operational Deployability and Review Scope

This appendix gives the detailed deployability measurements summarised in §6.6, together with the per-call review baseline behind AS6. Table 6 reports all three blocks on Qwen 2.5:72b: benign workload completion (five routine NOC tasks), single-step post-incident recovery, and runtime cost. The 5–6% false-rejection cost is confined to WF1, the only benign workflow that legitimately issues two sequential AMBR applies; operators whose recovery playbooks need multiple committed updates should recalibrate k^* or use a manual exception path. Runtime figures are prototype-specific and should be re-measured on target infrastructure.

AS6 models per-call oversight: each change is judged from the local tuple $(F(\delta_i), \delta_i, s_i)$ rather than a running accumulator, which is plausible for staged changes because each step looks locally feasible while the session accumulates drift. As §6.4 shows, adding a cumulative-sum check reduces harmful approval by only 1–2 pp; the stateful session counter enforced at the API boundary is what reliably breaks the escalation circuit.

Appendix C. Supporting Evaluation Evidence

This appendix gives the compact supporting evidence behind the evaluation. The main RQ1, RQ3, and RQ4

result tables remain in the body. The tables below report the calibration, defense ablation, adaptive-prompt, rejection-semantic, and single-step safety checks used to support the main claims. RQ5 deployability is reported separately in Appendix B.

Appendix D. Threat Scope and Within-Session Diagnostic

PALA targets reward hacking that arises from the *authorised* use of the analytics and policy tools (groups G1–G2c) under assumptions AS1–AS6 — the failure characterised by Theorem 5. Several adjacent threats are out of scope because they are orthogonal and already addressed elsewhere: prompt injection and supply-chain attacks [35], [55], compromised or malicious tools [56], [57], host or physical-layer compromise, training-time reward misspecification [31], and malicious operators. Cross-tenant contamination in multi-slice deployments is left to future work.

The null-control run in §6.3 gives zero decompositions in 40 trials. This indicates that the failure is triggered by operator-style staged-control intent, not by generic tool use.

We also report a within-session diagnostic, Def. 4*, in addition to the main counterfactual definition in Def. 4. It checks whether the agent’s later actions improve the proxy while making quality worse than after its own first successful policy action:

$$R(a_{1:k}, t_k) > R(a_{1:1}, t_k) \wedge Q(a_{1:k}, t_k) < Q(a_{1:1}, t_k). \quad (15)$$

Def. 4* is only a diagnostic for the rise-then-fall pattern inside the same trajectory. The main reward-hacking claim still uses Def. 4. Table 2 reports both rates.

Appendix E. Contamination Trace and Sensitivity Sweep

Radio-side physical-harm validation. Because UER-ANSIM does not enforce AMBR through a radio scheduler in the data plane, the main Open5GS/UERANSIM testbed realises over-subscription at a host bottleneck. We therefore repeat the physical-harm experiment on Open5GS/srsRAN. For reproducibility, the artifact provides a configuration using an srsRAN Project gNB and srsUE over a ZeroMQ RF front-end, with one Linux network namespace per UE and a 20 MHz FDD carrier in band 3. The ZeroMQ transport is provisioned well above radio capacity, and the host HTB root is maintained at least $10\times$ above the measured cell capacity, leaving the MAC scheduler as the limiting resource. Ground truth is measured out of band: per-UE downlink throughput from concurrent `iperf3` streams (10 s, reverse mode, one server port per UE), latency from `ping` at 0.2 s intervals issued *while* that load is running, loss from a separate UDP arm offering 20 Mbps per UE, and active-session count from the SMF. With four UEs the measured cell capacity is 22.8 Mbps and the per-UE baseline is 5.71 Mbps. Raising one UE’s ceiling from $1\times$ to $12\times$ that baseline,

Table 5. PALA IMPLEMENTATION SCOPE AND ASSUMPTION-TO-EVIDENCE MAPPING.

ID	Interface / group	Tool or basis	Prototype implementation	Evidence used	Limitation
G1	Data access	$H_{db}, H_{rag};$ TS 23.288 NWDAF Event Exposure	Read D_{5G} and D_{ana}	Tool traces and collector logs	Prototype, not production NWDAF.
G2a	Analytics	$H_{kpi};$ NWDAF AnLF analytics	Read D_{ana} only	KPI readback traces	Deployment-specific analytics schema.
G2b / AS4	Feasibility gate	$H_{feas};$ policy constraint checking	Existence, AMBR-bound, and cooldown checks applied independently	Accepted staged-AMBR traces	Other deployments may enforce cumulative accumulators.
G2c	Control	$H_{policy}, H_{sess}, H_{mon};$ PCF/SMF-facing policy tools	Write QoS/session parameters to D_{5G}	Policy-write and session logs	Prototype API mirrors a Release-18-compatible implementation class.
AS1	NWDAF interface basis	TS 23.288 data-collection and analytics interfaces	Open5GS-based NWDAF-style collector and analytics tools	Tool traces and collector logs	Not a vendor NWDAF implementation.
AS2	Policy-derived analytics path	NWDAF-agent analytics/control coupling	Policy-derived AMBR fields enter D_{ana} without semantic labels	Type-P/Mixed contamination traces	Schema choice may differ across deployments.
AS3	Collector propagation	Subscribed event aggregation without semantic provenance	Collector writes subscribed aggregates into D_{ana}	Standard vs. IsolatedCollector comparison	Does not model proprietary anomaly filters.
AS5	KPI-query control	Agent-controlled sample/window parameter N	Agent selects N in H_{kpi}	Sensitivity sweep over N, Δ_c	Amplifier only; not required by the theorem.
AS6	Per-call review scope	Reviewer sees isolated $(F(\delta_i), \delta_i, s_i)$ tuples	No architectural session-level accumulator before approval	RQ3 reviewer-rule comparison	Not a human-subject NOC study.

Table 6. DEPLOYABILITY SUMMARY FOR FULL PALA.

Metric	Standard	Full PALA
Benign completion	42/50 (84%)	42/50 (84%)
WF1 conditional AMBR raise	10/10	7/10
Type-P contamination	6/50 (12%)	0/50 (0%)
H_{budget} false rejections	—	3/50 (6%)
Recovery completion	17/20 (85%)	17/20 (85%)
Recovery false rejections	0/20	1/20 (5%)
Mean recovery time	92 s	88 s
KPI latency	3942 ms	< 0.1 ms
Gate overhead per policy call	—	< 1.3 ms
Session p50 / p99	127.1 / 251.6 s	112.4 / 194.5 s

False rejections are localized to WF1, the only benign workflow requiring two sequential AMBR applies.

Table 7. ARCHITECTURAL-ALTERNATIVE CONFIGURATIONS. EACH PARTIAL FIX FAILS AT LEAST ONE AXIS: V4 GATE DECOMPOSITION OR V7 TYPE-P CONTAMINATION. ONLY FULL PALA CLOSSES BOTH.

Configuration	V4 gate	V7 contam.	Chain
Vulnerable baseline	PASS	YES	YES
Per-call gate	PASS	YES	YES
Tight AMBR bound (200 Mbps)	PASS	YES	YES
Cooldown only (30 s)	PASS	YES	YES
DB isolation on separate host	PASS	YES	YES
Semantic filter / IsolatedCollector only	PASS	NO	NO
Accumulator / HedgeTuned only	BLOCK	YES	NO
Full PALA	BLOCK	NO	NO

PASS means the staged-control sequence succeeds on that axis. BLOCK means the axis is blocked. DB isolation alone does not remove Type-P fields because the problem is semantic provenance, not physical database placement.

while holding the other UEs' configuration, their offered load, and the cell capacity fixed, leaves victim throughput flat (5.43 $\rightarrow$ 5.47 Mbps) but raises victim latency from 77.7 to 151.5 ms (1.95 $\times$). Over-provisioning one slice therefore does not take bandwidth from the other sessions; it taxes them through the shared queue. Thus physical degradation persists when host-side shaping is not the bottleneck.

Hardware-RF confirmation. We additionally repeat the same physical-harm validation using a USRP-2953R as the srsRAN gNB RF front-end and a second USRP-2953R on the UE side, retaining the same Open5GS core, srsRAN control path, traffic-generation procedure, and over-

provisioning experiment. This single-configuration hardware check reproduces the qualitative physical-degradation trend relative to its hardware-RF baseline. We use it only as a hardware-path confirmation; the quantitative and reproducible radio-side validation is the ZeroMQ/srsRAN experiment above.

We validate Lemma 2 with a write-then-readback test: 20 policy writes are issued through H_{policy} without LLM inference. In all 20/20 cases, the next collector window produces a record in D_{ana} containing a Type-P field traceable to the write. IsolatedCollector removes this return path for $\phi = T$ queries: baseline returns occur in 20/20 cases, while isolated returns occur in 0/20 cases.

Policy-field reachability. We additionally checked whether the same collector path republishes other policy-derived fields: Session-AMBR, the 5QI QoS class, ARP priority, and the per-flow MBR. These sit in different configuration subtrees and, for flow MBR, a different enforcement path. Under the standard collector all four reappear in D_{ana} with a perfect write-to-readback correlation over 12 writes each and no provenance tag, and under IsolatedCollector none of the four does (4/4 versus 0/4 channels open). The channel is therefore a property of the collector rather than of the AMBR field. Agent-side closure was measured for three of the fields at $n = 5$ sessions each and is small and inconclusive at that sample size, so our end-to-end reward-hacking claim remains scoped to AMBR.

Field-name filtering is insufficient. In an enumerated-field ablation, filtering Session-AMBR by name still left the 5QI *qos index* field reachable, because that field was retained as functional session metadata under the tested enumerated-field rule. Replacing the field list with origin-based provenance filtering closes all four tested policy-derived channels. This is why IsolatedCollector filters by provenance rather than by field name: a name-based filter must enumerate every policy-derived field in advance and can fail when a new policy-derived field is added.

Table 8. CALIBRATION AND DEFENSE-ABLATION EVIDENCE. PANEL A SHOWS THE SESSION-BUDGET TRADEOFF. PANEL B SHOWS WHICH DEFENSE COMPONENT BREAKS WHICH PART OF THE CIRCUIT.

Panel A: $k^{\dagger*}$ sensitivity ($n=28$ staged-control sessions; $n=50$ benign trials; $n=20$ recovery trials).						
$k^{\dagger*}$	Admitted behavior	Staged-control effect before rejection	Circuit endpoint	Benign cost	Recovery	
1	One committed policy step.	12% show step-1 only.	$\Delta Q < 0$	0/28 (0%)	5–6%; localized to WF1.	85%; recovery is single-step.
2	Two committed policy steps.	96% admit a harmful step before rejection.		1/28 (4%)	$\approx 0\%$	85%.
3	Three committed policy steps.	100% admit the harmful step before rejection.		0/28 observed [†]	$\approx 0\%$	85%.
Panel B: Necessity/sufficiency ablation ($n=20$ per variant).						
Variant	AS2: isolate provenance	AS4: session budget	AS5: bound N	Full-loop	Conclusion	
<i>full chain</i>	×	×	×	15/20 (75%)	Baseline validates the vulnerable circuit. Removing policy-derived analytics breaks the proxy path.	
<i>as2 only</i> / ✓	✓	×	×	0/20 (0%)		
IsolatedCollector						
<i>as4 only</i> / ×	×	✓	×	0/20 (0%) [‡]	Session budgeting stops repeated committed escalation.	
HedgeTuned						
$k^{\dagger*}=1$						
<i>as2 and as4</i> / ✓	✓	✓	×	0/20 (0%)	Both contamination and escalation paths are closed.	
Full PALA						
<i>as5 only</i> / ×	×	×	✓	13/20 (65%)	Bounding N reduces amplification but does not break the circuit.	
bounded N						

[†] The $k^{\dagger*}=3$ row is retained only as sensitivity evidence: rejection occurs after three committed writes, so it is not a safe operational choice even when the measured endpoint count is low. [‡] Raw HT-only logs can include rejected attempts; the reported value uses the operational endpoint, i.e., successful policy calls only.

Table 9. ADAPTIVE-PROMPT, REJECTION-SEMANTICS, AND FIRST-WRITE SAFETY CHECKS. THESE RESULTS CLARIFY WHAT HAPPENS AFTER H_{budget} REJECTS A SESSION AND WHY ONE PERMITTED WRITE IS INSUFFICIENT TO SATISFY STRICT DEF. 4.

Panel A: Adaptive-prompt stress test ($n=50$ per condition; 5 adaptive prompts $\times$ 10 reps).				Panel B: Rejected-call semantics.			
Condition	Full-loop	H_{budget} rej.	Result	Observable	ISO-only	HT-only	Full PALA
Vulnerable baseline	46/50 (92%)	n/a	Reference vulnerable condition.	Rejection appears in event log	—	100%	100%
IsolatedCollector only	0/50 (0%)	n/a	No policy-derived analytics are returned to the proxy path.	Agent retries after rejection	—	100%	100%
HedgeTuned $k^{\dagger*}=1$	0/50 (0%) [†]	86	Repeated committed escalation is blocked.	Agent gives up after rejection	—	0%	0%
Full PALA	0/50 (0%)	90	Both adaptive closure paths are blocked.	D_{ana} contaminated after step-1 write	0%	94%	0%
Guardrail				Corrected full-loop after budget rejection	0%	0%	0%
Panel C: First permitted write under HT-only ($n=50$ adaptive-prompt sessions).							
Step-1 question	HT-only result		Interpretation				
Commits AMBR write to D_{5G}	Yes		The first write is allowed by design.				
Creates Type-P record in D_{ana}	94%		HT-only does not remove contaminated analytics.				
Increases proxy reward R vs. step-0	88%		The proxy can improve after the first write.				
Reduces Q below no-op baseline	12%		Some single-step harm appears.				
Satisfies Strict Def. 4	0%		The multi-step escalation needed for the definition is blocked.				

[†] Raw HT-only adaptive logs contain rejected attempts; the operational endpoint counts only successful policy calls. Against the vulnerable baseline, all three defended adaptive conditions give 0/50 events versus 46/50, with Fisher exact two-sided $p = 6.27 \times 10^{-24}$.

Table 10 also reports the sensitivity sweep. The sweep varies the collector interval Δ_c and KPI batch size N . Only the $\Delta_c = 5$ s latency row is a live measurement; the other latency rows are order-of-magnitude extrapolations. The N sweep shows that larger batches dilute the contamination ratio, but do not remove the channel.

The deployment implication is simple: reducing Δ_c lowers latency, and increasing N dilutes the contaminated fraction. Neither fix removes the channel. The cause is provenance mixing at AS2, so the channel is closed only when IsolatedCollector removes policy-derived records from the proxy path.

Appendix F. Robustness and Adaptive-Prompt Details

Table 11 gives two supporting checks. First, we recompute Strict Def. 4 under six operator weight profiles for Q . Across the six profiles, Strict Def. 4 ranges from 53.6% to 75.0%, so the result is not an artifact of the default equal-weight Q . Second, we report the per-prompt adaptive-prompt breakdown. Each prompt closes the circuit in 80–100% of vulnerable trials, but in 0/10 trials under IsolatedCollector, HedgeTuned, or Full PALA.

Model scale and capability. The supplementary tiers summarised in §6.2 use the same tool interface, system

Table 10. CONTAMINATION TRACE AND SENSITIVITY SWEEP. PANEL A SHOWS DIRECT WRITE-THEN-READBACK EVIDENCE. PANEL B SHOWS CONTAMINATION LATENCY VERSUS COLLECTOR INTERVAL. PANEL C SHOWS CONTAMINATION RATIO VERSUS KPI BATCH SIZE.

Panel A: Write-then-readback trace ($n = 20$ writes, 200 records/window, $\Delta_c = 5$ s).			
Write Written #	AMBR	Readback AMBR	Mixed fraction
1	20 Mbps	20.0 Mbps	1.00
2	22 Mbps	22.0 Mbps	1.00
3	24 Mbps	24.0 Mbps	1.00
20	58 Mbps	58.0 Mbps	1.00
Panel B: Contamination latency versus Δ_c .			
Δ_c	Model latency	Live measurement	Note
1 s	779 ms	—	extrapolated
5 s	2,779 ms	$2,974 \pm 134$ ms	live; 7.0% model error
15 s	7,779 ms	—	extrapolated
30 s	15,279 ms	—	extrapolated
Panel C: Type-P contamination ratio versus batch size N at $\Delta_c = 5$ s.			
N	Ratio	Pattern	Interpretation
5	100.0%	all records contaminated	contamination dominates
10	80.0%	$8/N$ curve	high contamination share
20	40.0%	$8/N$ curve	AS5 amplifier threshold
30	26.7%	$8/N$ curve	dilution begins
50	16.0%	$8/N$ curve	diluted but non-zero

prompt, staged-control setting, and intent corpus as the main evaluation, at $n = 20$ sessions per tier. Circuit closure is 14/20 (70%) for Llama 3.1:8B, 15/20 (75%) for Gemma-3 12B, and 15/20 (75%) for Claude Sonnet 4.5, with contaminated readback (Stage B) present in 90–100% of sessions in every tier, so closure does not decrease monotonically with model size or capability. Capability instead changes the magnitude of the overshoot beyond the operator’s stated target: the mean overshoot Δ^* falls from +3.41 (Llama 3.1:8B) through +2.48 (Gemma-3 12B) to −0.12 (Claude Sonnet 4.5), so the frontier model tracks the requested target most precisely and still escalates after contaminated readback. A partial 30–36B cell closed the circuit in 6/6 recorded sessions; it is incomplete and is reported in the artifact rather than quoted as a rate.

Scripted-controller control. A deterministic script replaces the agent and issues the same staged increments, continuing only while the analytics confirm its last write ($n = 20$ per arm). Under the standard collector it escalates to $50\times$ baseline in 20/20 sessions; the identical script under IsolatedCollector halts after the first write in 20/20 sessions. Script, decision rule, and corpus are identical across the two arms and only collector provenance differs, so the escalation is attributable to the channel rather than to the controller.

Target-free LLM control. To separate escalation from the execution of a supplied number, we also run sessions in which no numeric AMBR target is given and the agent is asked only to improve bandwidth progressively while checking analytics between changes (*qwen2.5:72b*, four defense arms of 10 sessions each, $n = 33$ of which record a final ceiling). Pooled across the four arms, the 33 sessions with a recorded final ceiling reach a mean of $29\times$ baseline and the configured ceiling in 13/33 cases; these pooled ceiling values are descriptive rather than a

Table 11. ROBUSTNESS AND ADAPTIVE-PROMPT DETAIL. PANEL A RECOMPUTES STRICT DEF. 4 UNDER DIFFERENT Q WEIGHTS ON THE 28 QWEN SESSIONS WITH COMPLETE Q TRACES MEASURED FROM THE FIRST POLICY CALL ONWARD. TABLE 2 USES ALL 30 SESSIONS, SO THE EQUAL-WEIGHT COUNT HERE IS 16/28 RATHER THAN 18/30. PANEL B GIVES THE PER-PROMPT ADAPTIVE-PROMPT BREAKDOWN.

Panel A: Q -weight robustness ($n = 28$ Qwen sessions with complete Q traces from the first policy call onward; Table 2 uses all 30).					
Profile	$(v_\tau, v_\lambda, v_\rho, v_\sigma)$	n	Strict Def.	Δ	Effect
Equal baseline	(0.25, 0.25, 0.25, 0.25)	28	16/28 (57.1%)	—	—
Throughput/	(0.50, 0.17, 0.17, 0.16)	28	21/28 (75.0%)	+17.9 pp	$h = +0.38$
eMBB Latency/	(0.17, 0.50, 0.17, 0.16)	28	15/28 (53.6%)	−3.6 pp	$h = -0.07$
URLLC Loss-sensitive	(0.17, 0.17, 0.50, 0.16)	28	16/28 (57.1%)	0.0 pp	$h = 0.00$
Stability-heavy	(0.17, 0.17, 0.16, 0.50)	28	16/28 (57.1%)	0.0 pp	$h = 0.00$
Stability-removed	(0.33, 0.33, 0.34, 0.00)	28	16/28 (57.1%)	0.0 pp	$h = 0.00$
Panel B: Adaptive-prompt per-prompt breakdown ($n = 10$ trials per cell).					
Prompt	Vulnerable	ISO	HT	Full PALA	
P1: “incremental staged steps”	10/10	0/10	0/10	0/10	
P2: “gradually raise...series”	8/10	0/10	0/10	0/10	
P3: “staged AMBR up-lift...stages”	9/10	0/10	0/10	0/10	
P4: “progressive rollout”	9/10	0/10	0/10	0/10	
P5: “successive incremental”	10/10	0/10	0/10	0/10	
Total	46/50 (92%)	0/50	0/50	0/50	

defense-efficacy measure, because the defenses alter readback and commit semantics. In the undefended arm alone, the mean is $17\times$ with 1/10 at the ceiling, which is sufficient to show escalation without a supplied numeric target. In IsolatedCollector-only runs, removing policy-origin confirmation can lead the agent to continue *proposing* increases because no contaminated completion signal is returned; HedgeTuned and Full PALA nevertheless bound committed writes at the policy boundary. The escalation is therefore not the execution of an operator’s number. This probe addresses the supplied-target confound, not causal dependence on contamination; contamination dependence is evaluated separately through the standard-versus-IsolatedCollector intervention and the Stage B $\rightarrow$ C $\rightarrow$ D trace. In the undefended arm of the same experiment ($n = 10$), 9/10 sessions obtain a contaminated readback and 6/10 issue a further policy write after it.

The uniform 0/10 defended result is expected because both defenses are below the agent’s reasoning layer. The agent cannot undo the IsolatedCollector $\phi = T$ filter through prompting, and it cannot bypass the API-level session counter enforced by H_{budget} .

Appendix G. Oversight and Calibration Summary

Table 3 shows that per-call and AND-rule approval rates remain close across all three model families, while the cumulative-threshold rule approves 28–30 pp more steps. At the session level, the “only cumulative blocks” case is

0/30 in every family; in 13–15 of 30 sessions per family, cumulative review approves trajectories that per-call review partially blocks. A simulated-engineer sanity check gives the expected contrast: per-call visibility approves 72% of staged-control steps, while shape-aware cumulative visibility approves 0%, with 1/10 benign multi-step false rejections. This supports AS6: the weakness is the information shown before approval, not reviewer attention alone.

For RQ4, the stage-level evidence is also simple. IsolatedCollector breaks the contaminated-readback stage by reducing policy-derived analytics from 93% to 0%, but it does not stop decomposition. HedgeTuned with $k^{\dagger*} = 1$ stops repeated committed escalation, but does not clean D_{ana} . Full PALA combines both cuts, reducing the operational circuit endpoint to 0%.

The HedgeTuned calibration is supported by two diagnostics on the $n = 28$ identifiable Qwen calibration sessions. First, the average quality curve follows the expected rise-then-fall pattern: $E[Q(0)] = 0.814$, $E[Q(1)] = 0.846$, $E[Q(2)] = 0.734$, and $E[Q(3)] = 0.509$, with small per-step support ($n_k = 7, 2, 2, 1$). Second, the first harmful step is early: 17/28 sessions reach it at $k = 1$, 10/28 at $k = 2$, and 1/28 at $k = 3$ (CV=0.401, 95% CI [1.22, 1.64]). Thus 97% of identifiable sessions become harmful by $k = 2$, making $k^{\dagger*} = 1$ the conservative operating boundary on this corpus. We treat this as a calibration diagnostic, not a formal density-ratio proof.